

MODULE 1

Introduction to Machine Learning

Machine Learning (ML) is a promising and flourishing field. It can enable top management of an organization to extract the knowledge from the data stored in various archives of the business organizations to facilitate decision making. Such decisions can be useful for organizations to design new products, improve business processes, and to develop decision support systems.

Learning Objectives

- Explore the basics of machine learning
- Introduce types of machine learning
- Provide an overview of machine learning tasks
- State the components of the machine learning algorithm
- Explore the machine learning process
- Survey some machine learning applications

1.1 NEED FOR MACHINE LEARNING

Business organizations use huge amount of data for their daily activities. Earlier, the full potential of this data was not utilized due to two reasons. One reason was data being scattered across different archive systems and organizations not being able to integrate these sources fully. Secondly, the lack of awareness about software tools that could help to unearth the useful information from data. Not anymore! Business organizations have now started to use the latest technology, machine learning, for this purpose.

Machine learning has become so popular because of three reasons:

1. High volume of available data to manage: Big companies such as Facebook, Twitter, and YouTube generate huge amount of data that grows at a phenomenal rate. It is estimated that the data approximately gets doubled every year.

2. Second reason is that the cost of storage has reduced. The hardware cost has also dropped. Therefore, it is easier now to capture, process, store, distribute, and transmit the digital information.
3. Third reason for popularity of machine learning is the availability of complex algorithms now. Especially with the advent of deep learning, many algorithms are available for machine learning.

With the popularity and ready adaption of machine learning by business organizations, it has become a dominant technology trend now. Before starting the machine learning journey, let us establish these terms - data, information, knowledge, intelligence, and wisdom. A knowledge pyramid is shown in Figure 1.1.

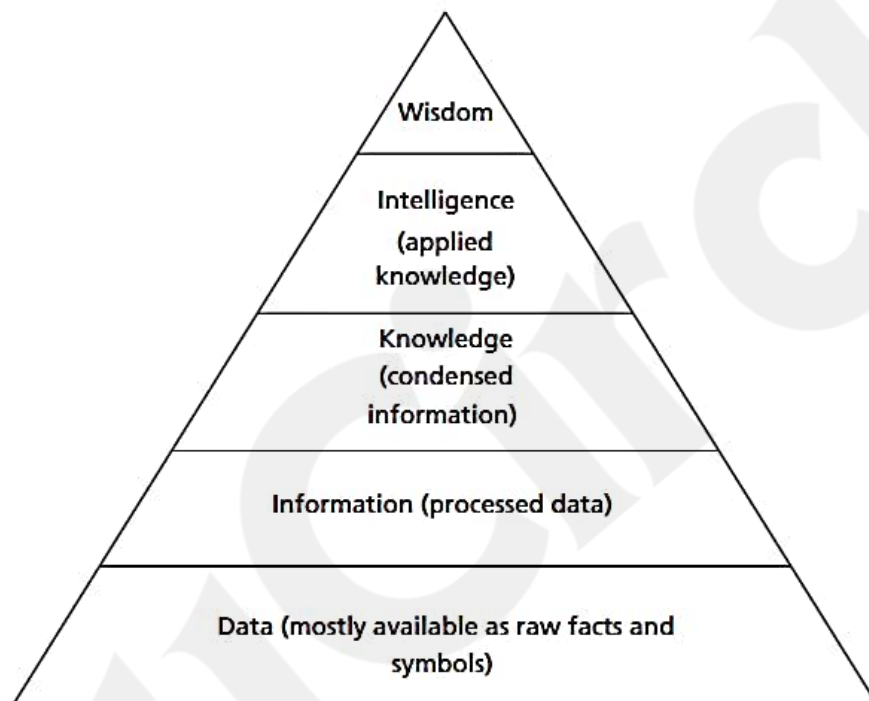


Figure 1.1: The Knowledge Pyramid

What is data? All facts are data. Data can be numbers or text that can be processed by a computer. Today, organizations are accumulating vast and growing amounts of data with data sources such as flat files, databases, or data warehouses in different storage formats.

Processed data is called information. This includes patterns, associations, or relationships among data. For example, sales data can be analyzed to extract information like which is the fast selling product. Condensed information is called knowledge. For example, the historical patterns and future trends obtained in the above sales data can be called knowledge. Unless knowledge is extracted, data is of no use. Similarly, knowledge is not useful unless it is put into action. Intelligence is the applied knowledge for actions. An actionable form of knowledge is called intelligence. Computer systems have been successful till this stage. The ultimate objective of knowledge pyramid is wisdom that represents the maturity of mind that is, so far, exhibited only by humans.

Here comes the need for machine learning. The objective of machine learning is to process these archival data for organizations to take better decisions to design new products, improve the business processes, and to develop effective decision support systems.

1.2 MACHINE LEARNING EXPLAINED

Machine learning is an important sub-branch of Artificial Intelligence (AI). A frequently quoted definition of machine learning was by Arthur Samuel, one of the pioneers of Artificial Intelligence. He stated that “*Machine learning is the field of study that gives the computers ability to learn without being explicitly programmed.*”

The key to this definition is that the systems should learn by itself without explicit programming. How is it possible? It is widely known that to perform a computation, one needs to write programs that teach the computers how to do that computation.

In conventional programming, after understanding the problem, a detailed design of the program such as a flowchart or an algorithm needs to be created and converted into programs using a suitable programming language. This approach could be difficult for many real-world problems such as puzzles, games, and complex image recognition applications. Initially, artificial intelligence aims to understand these problems and develop general purpose rules manually. Then, these rules are formulated into logic and implemented in a program to create intelligent systems. This idea of developing intelligent systems by using logic and reasoning by converting an expert’s knowledge into a set of rules and programs is called an expert system. An expert system like MYCIN was designed for medical diagnosis after converting the expert knowledge of many doctors into a system. However, this approach did not progress much as programs lacked real intelligence. The word MYCIN is derived from the fact that most of the antibiotics’ names end with ‘mycin’.

The above approach was impractical in many domains as programs still depended on human expertise and hence did not truly exhibit intelligence. Then, the momentum shifted to machine learning in the form of data driven systems. The focus of AI is to develop intelligent systems by using data-driven approach, where data is used as an input to develop intelligent models. The models can then be used to predict new inputs. Thus, the aim of machine learning is to learn a model or set of rules from the given dataset automatically so that it can predict the unknown data correctly.

As humans take decisions based on an experience, computers make models based on extracted patterns in the input data and then use these data-filled models for prediction and to take decisions. For computers, the learnt model is equivalent to human experience. This is shown in Figure 1.2.

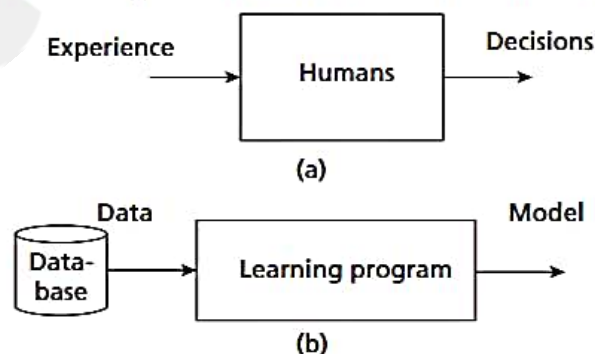


Figure 1.2: (a) A Learning System for Humans (b) A Learning System for Machine Learning

Often, the quality of data determines the quality of experience and, therefore, the quality of the learning system. In statistical learning, the relationship between the input x and output y is

modeled as a function in the form $y = f(x)$. Here, f is the learning function that maps the input x to output y . Learning of function f is the crucial aspect of forming a model in statistical learning. In machine learning, this is simply called mapping of input to output.

The learning program summarizes the raw data in a model. Formally stated, a model is an explicit description of patterns within the data in the form of:

1. Mathematical equation
2. Relational diagrams like trees/graphs
3. Logical if/else rules, or
4. Groupings called clusters

In summary, a model can be a formula, procedure or representation that can generate data decisions. The difference between pattern and model is that the former is local and applicable only to certain attributes but the latter is global and fits the entire dataset. For example, a model can be helpful to examine whether a given email is spam or not. The point is that the model is generated automatically from the given data.

Another pioneer of AI, Tom Mitchell's definition of machine learning states that, "*A computer program is said to learn from experience E , with respect to task T and some performance measure P , if its performance on T measured by P improves with experience E .*" The important components of this definition are experience E , task T , and performance measure P .

For example, the task T could be detecting an object in an image. The machine can gain the knowledge of object using training dataset of thousands of images. This is called experience E . So, the focus is to use this experience E for this task of object detection T . The ability of the system to detect the object is measured by performance measures like precision and recall. Based on the performance measures, course correction can be done to improve the performance of the system.

Models of computer systems are equivalent to human experience. Experience is based on data. Humans gain experience by various means. They gain knowledge by rote learning. They observe others and imitate it. Humans gain a lot of knowledge from teachers and books. We learn many things by trial and error. Once the knowledge is gained, when a new problem is encountered, humans search for similar past situations and then formulate the heuristics and use that for prediction. But, in systems, experience is gathered by these steps:

1. Collection of data
2. Once data is gathered, abstract concepts are formed out of that data. Abstraction is used to generate concepts. This is equivalent to humans' idea of objects, for example, we have some idea about how an elephant looks like.
3. Generalization converts the abstraction into an actionable form of intelligence. It can be viewed as ordering of all possible concepts. So, generalization involves ranking of concepts, inferencing from them and formation of heuristics, an actionable aspect of intelligence. Heuristics are educated guesses for all tasks. For example, if one runs or encounters a danger, it is the resultant of human experience or his heuristics formation. In machines, it happens the same way.
4. Heuristics normally works! But, occasionally, it may fail too. It is not the fault of heuristics as it is just a 'rule of thumb'. The course correction is done by taking evaluation measures. Evaluation checks the thoroughness of the models and to-do course correction, if necessary, to generate better formulations.

1.3 MACHINE LEARNING IN RELATION TO OTHER FIELDS

Machine learning uses the concepts of Artificial Intelligence, Data Science, and Statistics primarily. It is the resultant of combined ideas of diverse fields.

1.3.1 Machine Learning and Artificial Intelligence

Machine learning is an important branch of AI, which is a much broader subject. The aim of AI is to develop intelligent agents. An agent can be a robot, humans, or any autonomous systems. Initially, the idea of AI was ambitious, that is, to develop intelligent systems like human beings. The focus was on logic and logical inferences. It had seen many ups and downs. These down periods were called AI winters.

The resurgence in AI happened due to development of data driven systems. The aim is to find relations and regularities present in the data. Machine learning is the subbranch of AI, whose aim is to extract the patterns for prediction. It is a broad field that includes learning from examples and other areas like reinforcement learning. The relationship of AI and machine learning is shown in Figure 1.3. The model can take an unknown instance and generate results.

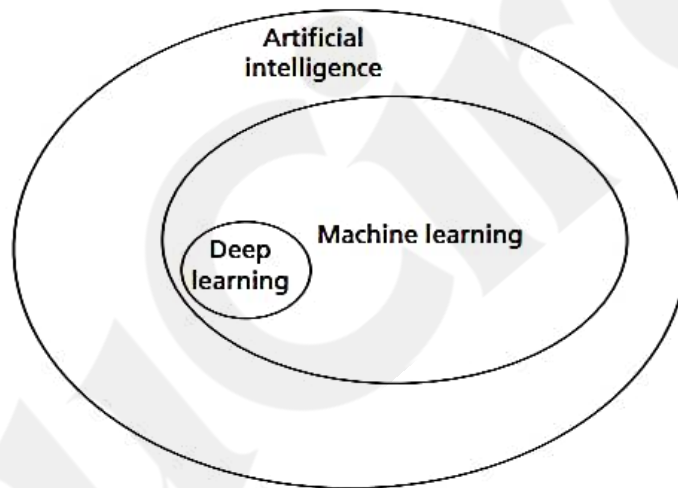


Figure 1.3: Relationship of AI with Machine Learning

Deep learning is a subbranch of machine learning. In deep learning, the models are constructed using neural network technology. Neural networks are based on the human neuron models. Many neurons form a network connected with the activation functions that trigger further neurons to perform tasks.

1.3.2 Machine Learning, Data Science, Data Mining, and Data Analytics

Data science is an 'Umbrella' term that encompasses many fields. Machine learning starts with data. Therefore, data science and machine learning are interlinked. Machine learning is a branch of data science. Data science deals with gathering of data for analysis. It is a broad field that includes:

Big Data Data science concerns about collection of data. Big data is a field of data science that deals with data's following characteristics:

1. Volume: Huge amount of data is generated by big companies like Facebook, Twitter, YouTube.
2. Variety: Data is available in variety of forms like images, videos, and in different formats.
3. Velocity: It refers to the speed at which the data is generated and processed.

Big data is used by many machine learning algorithms for applications such as language translation and image recognition. Big data influences the growth of subjects like Deep learning. Deep learning is a branch of machine learning that deals with constructing models using neural networks.

Data Mining Data mining's original genesis is in the business. Like while mining the earth one gets into precious resources, it is often believed that unearthing of the data produces hidden information that otherwise would have eluded the attention of the management. Nowadays, many consider that data mining and machine learning are same. There is no difference between these fields except that data mining aims to extract the hidden patterns that are present in the data, whereas, machine learning aims to use it for prediction.

Data Analytics Another branch of data science is data analytics. It aims to extract useful knowledge from crude data. There are different types of analytics. Predictive data analytics is used for making predictions. Machine learning is closely related to this branch of analytics and shares almost all algorithms.

Pattern Recognition It is an engineering field. It uses machine learning algorithms to extract the features for pattern analysis and pattern classification. One can view pattern recognition as a specific application of machine learning.

These relations are summarized in Figure 1.4.

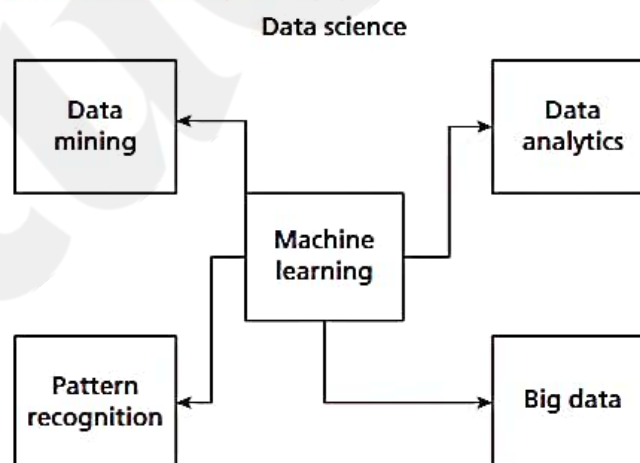


Figure 1.4: Relationship of Machine Learning with Other Major Fields

1.3.3 Machine Learning and Statistics

Statistics is a branch of mathematics that has a solid theoretical foundation regarding statistical learning. Like machine learning (ML), it can learn from data. But the difference between statistics and ML is that statistical methods look for regularity in data called patterns. Initially, statistics sets a hypothesis and performs experiments to verify and validate the hypothesis in order to find relationships among data.

Statistics requires knowledge of the statistical procedures and the guidance of a good statistician. It is mathematics intensive and models are often complicated equations and involve many assumptions. Statistical methods are developed in relation to the data being analysed. In addition, statistical methods are coherent and rigorous. It has strong theoretical foundations and interpretations that require a strong statistical knowledge.

Machine learning, comparatively, has less assumptions and requires less statistical knowledge. But, it often requires interaction with various tools to automate the process of learning.

Nevertheless, there is a school of thought that machine learning is just the latest version of 'old Statistics' and hence this relationship should be recognized.

1.4 TYPES OF MACHINE LEARNING

What does the word 'learn' mean? Learning, like adaptation, occurs as the result of interaction of the program with its environment. It can be compared with the interaction between a teacher and a student. There are four types of machine learning as shown in Figure 1.5.

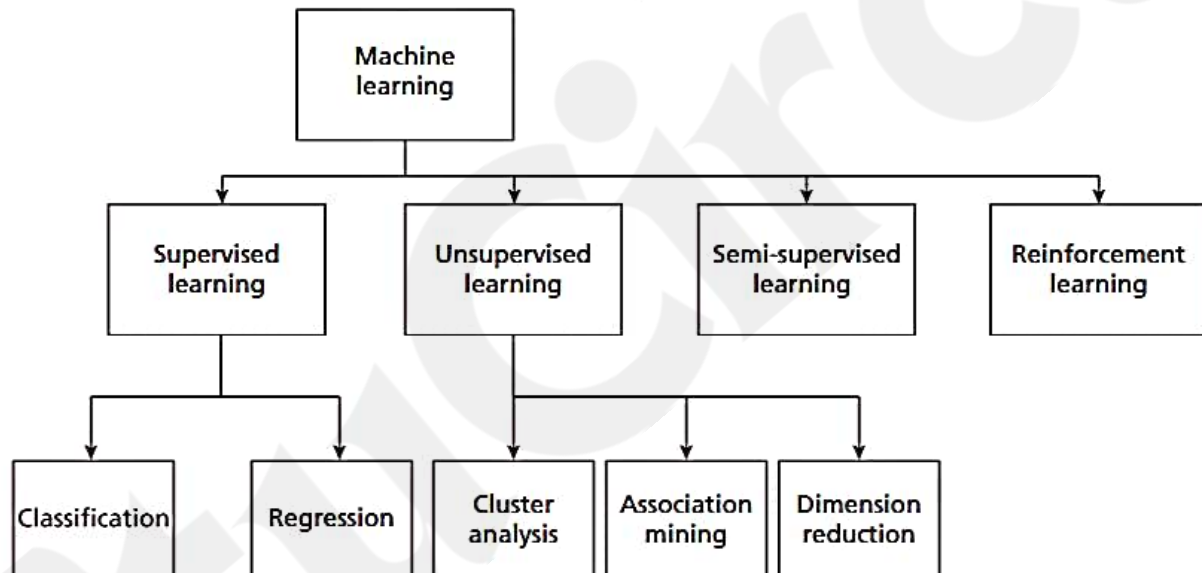


Figure 1.5: Types of Machine Learning

Before discussing the types of learning, it is necessary to discuss about data.

Labelled and Unlabelled Data Data is a raw fact. Normally, data is represented in the form of a table. Data also can be referred to as a data point, sample, or an example. Each row of the table represents a data point. Features are attributes or characteristics of an object. Normally, the columns of the table are attributes. Out of all attributes, one attribute is important and is called a label. Label is the feature that we aim to predict. Thus, there are two types of data – labelled and unlabelled.

Labelled Data To illustrate labelled data, let us take one example dataset called Iris flower dataset or Fisher's Iris dataset. The dataset has 50 samples of Iris – with four attributes, length and width of sepals and petals. The target variable is called class. There are three classes – Iris setosa, Iris virginica, and Iris versicolor.

The partial data of Iris dataset is shown in Table 1.1.

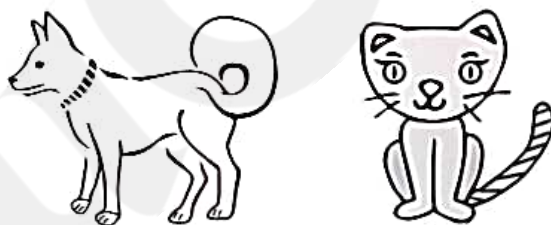
Table 1.1: Iris Flower Dataset

S.No.	Length of Petal	Width of Petal	Length of Sepal	Width of Sepal	Class
1.	5.5	4.2	1.4	0.2	Setosa
2.	7	3.2	4.7	1.4	Versicolor
3.	7.3	2.9	6.3	1.8	Virginica

A dataset need not be always numbers. It can be images or video frames. Deep neural networks can handle images with labels. In the following Figure 1.6, the deep neural network takes images of dogs and cats with labels for classification.

Input	Label
	dog
	Cat

(a)



(b)

Figure 1.6: (a) Labelled Dataset (b) Unlabelled Dataset

In unlabelled data, there are no labels in the dataset.

1.4.1 Supervised Learning

Supervised algorithms use labelled dataset. As the name suggests, there is a supervisor or teacher component in supervised learning. A supervisor provides labelled data so that the model is constructed and generates test data.

In supervised learning algorithms, learning takes place in two stages. In layman terms, during the first stage, the teacher communicates the information to the student that the student is supposed to master. The student receives the information and understands it. During this stage, the teacher has no knowledge of whether the information is grasped by the student.

This leads to the second stage of learning. The teacher then asks the student a set of questions to find out how much information has been grasped by the student. Based on these questions,

the student is tested, and the teacher informs the student about his assessment. This kind of learning is typically called supervised learning.

Supervised learning has two methods:

1. Classification
2. Regression

Classification

Classification is a supervised learning method. The input attributes of the classification algorithms are called independent variables. The target attribute is called label or dependent variable. The relationship between the input and target variable is represented in the form of a structure which is called a classification model. So, the focus of classification is to predict the 'label' that is in a discrete form (a value from the set of finite values). An example is shown in Figure 1.7 where a classification algorithm takes a set of labelled data images such as dogs and cats to construct a model that can later be used to classify an unknown test image data.

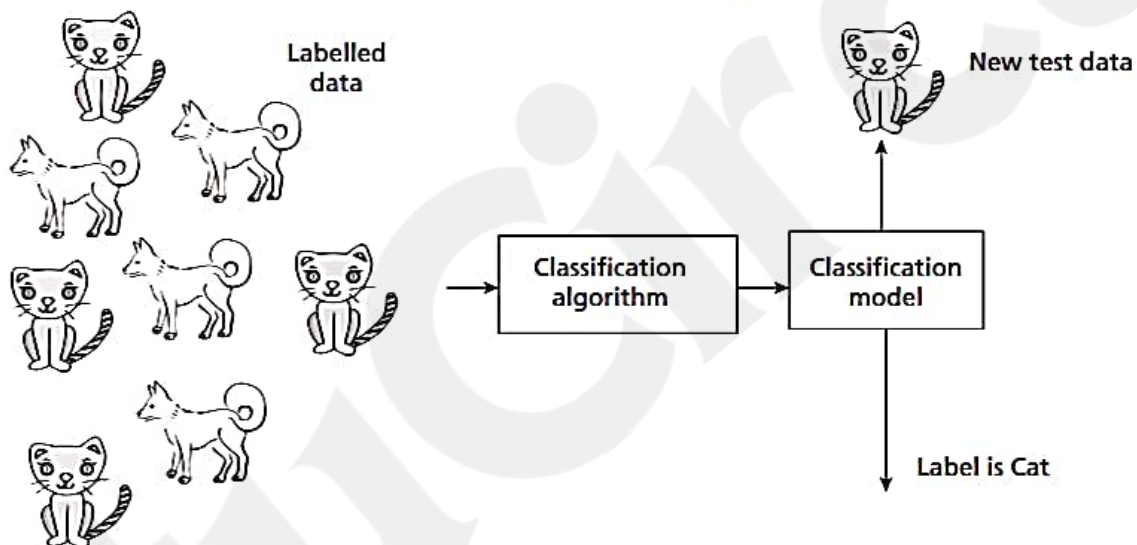


Figure 1.7: An Example Classification System

In classification, learning takes place in two stages. During the first stage, called training stage, the learning algorithm takes a labelled dataset and starts learning. After the training set, samples are processed and the model is generated. In the second stage, the constructed model is tested with test or unknown sample and assigned a label. This is the classification process.

This is illustrated in the above Figure 1.7. Initially, the classification learning algorithm learns with the collection of labelled data and constructs the model. Then, a test case is selected, and the model assigns a label.

Similarly, in the case of Iris dataset, if the test is given as (6.3, 2.9, 5.6, 1.8, ?), the classification will generate the label for this. This is called classification. One of the examples of classification is – Image recognition, which includes classification of diseases like cancer, classification of plants, etc.

The classification models can be categorized based on the implementation technology like decision trees, probabilistic methods, distance measures, and soft computing methods. Classification models can also be classified as generative models and discriminative models. Generative models deal with the process of data generation and its distribution. Probabilistic models are examples of

generative models. Discriminative models do not care about the generation of data. Instead, they simply concentrate on classifying the given data.

Some of the key algorithms of classification are:

- Decision Tree
- Random Forest
- Support Vector Machines
- Naïve Bayes
- Artificial Neural Network and Deep Learning networks like CNN

Regression Models

Regression models, unlike classification algorithms, predict continuous variables like price. In other words, it is a number. A fitted regression model is shown in Figure 1.8 for a dataset that represent weeks input x and product sales y .

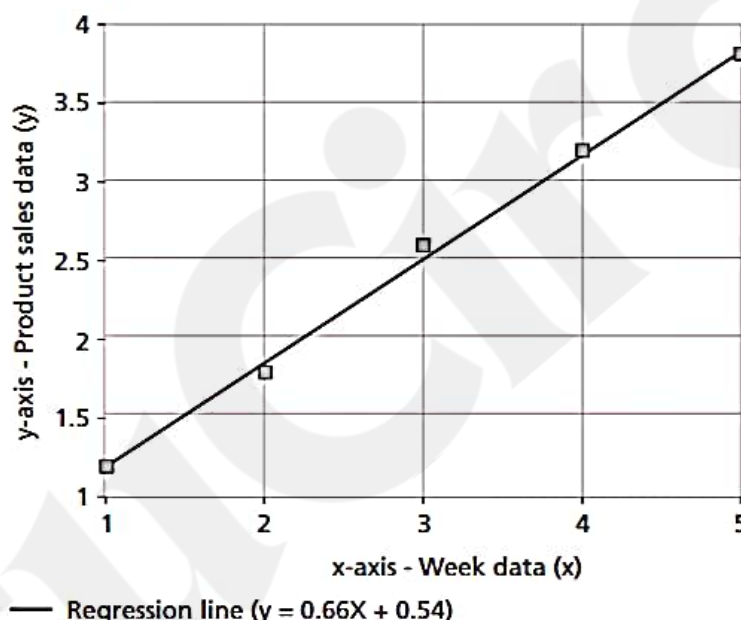


Figure 1.8: A Regression Model of the Form $y = ax + b$

The regression model takes input x and generates a model in the form of a fitted line of the form $y = f(x)$. Here, x is the independent variable that may be one or more attributes and y is the dependent variable. In Figure 1.8, linear regression takes the training set and tries to fit it with a line – product sales = $0.66 \times \text{Week} + 0.54$. Here, 0.66 and 0.54 are all regression coefficients that are learnt from data. The advantage of this model is that prediction for product sales (y) can be made for unknown week data (x). For example, the prediction for unknown eighth week can be made by substituting x as 8 in that regression formula to get y .

One of the most important regression algorithms is linear regression that is explained in the next section.

Both regression and classification models are supervised algorithms. Both have a supervisor and the concepts of training and testing are applicable to both. What is the difference between classification and regression models? The main difference is that regression models predict continuous variables such as product price, while classification concentrates on assigning labels such as class.

1.4.2 Unsupervised Learning

The second kind of learning is by self-instruction. As the name suggests, there are no supervisor or teacher components. In the absence of a supervisor or teacher, self-instruction is the most common kind of learning process. This process of self-instruction is based on the concept of trial and error.

Here, the program is supplied with objects, but no labels are defined. The algorithm itself observes the examples and recognizes patterns based on the principles of grouping. Grouping is done in ways that similar objects form the same group.

Cluster analysis and Dimensional reduction algorithms are examples of unsupervised algorithms.

Cluster Analysis

Cluster analysis is an example of unsupervised learning. It aims to group objects into disjoint clusters or groups. Cluster analysis clusters objects based on its attributes. All the data objects of the partitions are similar in some aspect and vary from the data objects in the other partitions significantly.

Some of the examples of clustering processes are — segmentation of a region of interest in an image, detection of abnormal growth in a medical image, and determining clusters of signatures in a gene database.

An example of clustering scheme is shown in Figure 1.9 where the clustering algorithm takes a set of dogs and cats images and groups it as two clusters—dogs and cats. It can be observed that the samples belonging to a cluster are similar and samples are different radically across clusters.

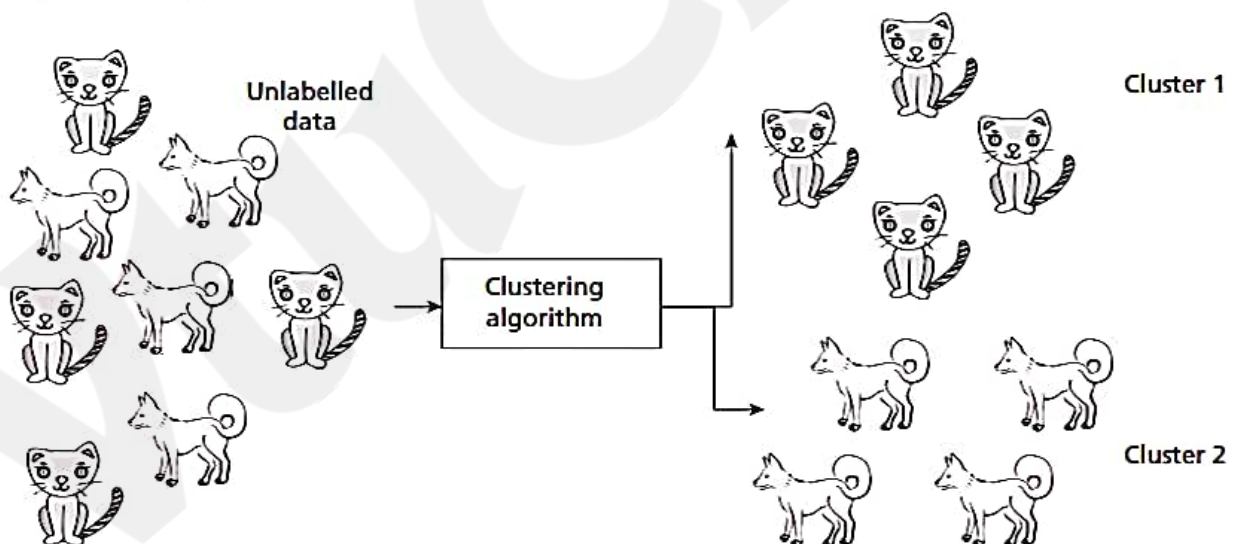


Figure 1.9: An Example Clustering Scheme

Some of the key clustering algorithms are:

- k-means algorithm
- Hierarchical algorithms

Dimensionality Reduction

Dimensionality reduction algorithms are examples of unsupervised algorithms. It takes a higher dimension data as input and outputs the data in lower dimension by taking advantage of the variance of the data. It is a task of reducing the dataset with few features without losing the generality.

The differences between supervised and unsupervised learning are listed in the following Table 1.2.

Table 1.2: Differences between Supervised and Unsupervised Learning

S.No.	Supervised Learning	Unsupervised Learning
1.	There is a supervisor component	No supervisor component
2.	Uses Labelled data	Uses Unlabelled data
3.	Assigns categories or labels	Performs grouping process such that similar objects will be in one cluster

1.4.3 Semi-supervised Learning

There are circumstances where the dataset has a huge collection of unlabelled data and some labelled data. Labelling is a costly process and difficult to perform by the humans. Semi-supervised algorithms use unlabelled data by assigning a pseudo-label. Then, the labelled and pseudo-labelled dataset can be combined.

1.4.4 Reinforcement Learning

Reinforcement learning mimics human beings. Like human beings use ears and eyes to perceive the world and take actions, reinforcement learning allows the agent to interact with the environment to get rewards. The agent can be human, animal, robot, or any independent program. The rewards enable the agent to gain experience. The agent aims to maximize the reward.

The reward can be positive or negative (Punishment). When the rewards are more, the behavior gets reinforced and learning becomes possible.

Consider the following example of a Grid game as shown in Figure 1.10.

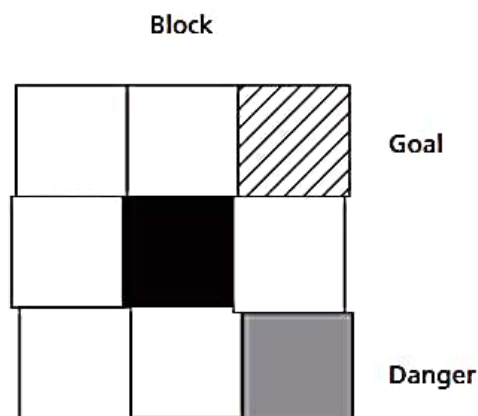


Figure 1.10: A Grid game

In this grid game, the gray tile indicates the danger, black is a block, and the tile with diagonal lines is the goal. The aim is to start, say from bottom-left grid, using the actions left, right, top and bottom to reach the goal state.

To solve this sort of problem, there is no data. The agent interacts with the environment to get experience. In the above case, the agent tries to create a model by simulating many paths and finding rewarding paths. This experience helps in constructing a model.

It can be said in summary, compared to supervised learning, there is no supervisor or labelled dataset. Many sequential decisions need to be taken to reach the final decision. Therefore, reinforcement algorithms are reward-based, goal-oriented algorithms.

Scan for information on 'Important Machine Learning Algorithms'



1.5 CHALLENGES OF MACHINE LEARNING

What are the challenges of machine learning? Let us discuss about them now.

Problems that can be Dealt with Machine Learning

Computers are better than humans in performing tasks like computation. For example, while calculating the square root of large numbers, an average human may blink but computers can display the result in seconds. Computers can play games like chess, GO, and even beat professional players of that game.

However, humans are better than computers in many aspects like recognition. But, deep learning systems challenge human beings in this aspect as well. Machines can recognize human faces in a second. Still, there are tasks where humans are better as machine learning systems still require quality data for model construction. The quality of a learning system depends on the quality of data. This is a challenge. Some of the challenges are listed below:

1. Problems – Machine learning can deal with the 'well-posed' problems where specifications are complete and available. Computers cannot solve 'ill-posed' problems.

Consider one simple example (shown in Table 1.3):

Table 1.3: An Example

Input (x_1, x_2)	Output (y)
1, 1	1
2, 1	2
3, 1	3
4, 1	4
5, 1	5

Can a model for this test data be multiplication? That is, $y = x_1 \times x_2$. Well! It is true! But, this is equally true that y may be $y = x_1 \div x_2$, or $y = x_1^{x_2}$. So, there are three functions that fit the data. This means that the problem is ill-posed. To solve this problem, one needs more example to check the model. Puzzles and games that do not have sufficient specification may become an ill-posed problem and scientific computation has many ill-posed problems.

2. Huge data – This is a primary requirement of machine learning. Availability of a quality data is a challenge. A quality data means it should be large and should not have data problems such as missing data or incorrect data.
3. High computation power – With the availability of Big Data, the computational resource requirement has also increased. Systems with *Graphics Processing Unit* (GPU) or even *Tensor Processing Unit* (TPU) are required to execute machine learning algorithms. Also, machine learning tasks have become complex and hence time complexity has increased, and that can be solved only with high computing power.
4. Complexity of the algorithms – The selection of algorithms, describing the algorithms, application of algorithms to solve machine learning task, and comparison of algorithms have become necessary for machine learning or data scientists now. Algorithms have become a big topic of discussion and it is a challenge for machine learning professionals to design, select, and evaluate optimal algorithms.
5. Bias/Variance – Variance is the error of the model. This leads to a problem called bias/variance tradeoff. A model that fits the training data correctly but fails for test data, in general lacks generalization, is called overfitting. The reverse problem is called underfitting where the model fails for training data but has good generalization. Overfitting and underfitting are great challenges for machine learning algorithms.

1.6 MACHINE LEARNING PROCESS

The emerging process model for the data mining solutions for business organizations is CRISP-DM. Since machine learning is like data mining, except for the aim, this process can be used for machine learning. CRISP-DM stands for Cross Industry Standard Process – Data Mining. This process involves six steps. The steps are listed below in Figure 1.11.

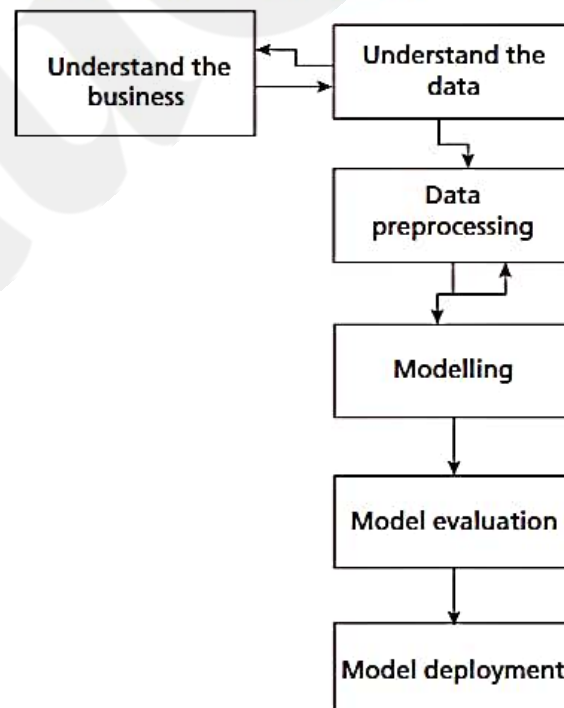


Figure 1.11: A Machine Learning/Data Mining Process

1. **Understanding the business** – This step involves understanding the objectives and requirements of the business organization. Generally, a single data mining algorithm is enough for giving the solution. This step also involves the formulation of the problem statement for the data mining process.
2. **Understanding the data** – It involves the steps like data collection, study of the characteristics of the data, formulation of hypothesis, and matching of patterns to the selected hypothesis.
3. **Preparation of data** – This step involves producing the final dataset by cleaning the raw data and preparation of data for the data mining process. The missing values may cause problems during both training and testing phases. Missing data forces classifiers to produce inaccurate results. This is a perennial problem for the classification models. Hence, suitable strategies should be adopted to handle the missing data.
4. **Modelling** – This step plays a role in the application of data mining algorithm for the data to obtain a model or pattern.
5. **Evaluate** – This step involves the evaluation of the data mining results using statistical analysis and visualization methods. The performance of the classifier is determined by evaluating the accuracy of the classifier. The process of classification is a fuzzy issue. For example, classification of emails requires extensive domain knowledge and requires domain experts. Hence, performance of the classifier is very crucial.
6. **Deployment** – This step involves the deployment of results of the data mining algorithm to improve the existing process or for a new situation.

1.7 MACHINE LEARNING APPLICATIONS

Machine Learning technologies are used widely now in different domains. Machine learning applications are everywhere! One encounters many machine learning applications in the day-to-day life. Some applications are listed below:

1. **Sentiment analysis** – This is an application of natural language processing (NLP) where the words of documents are converted to sentiments like happy, sad, and angry which are captured by emoticons effectively. For movie reviews or product reviews, five stars or one star are automatically attached using sentiment analysis programs.
2. **Recommendation systems** – These are systems that make personalized purchases possible. For example, Amazon recommends users to find related books or books bought by people who have the same taste like you, and Netflix suggests shows or related movies of your taste. The recommendation systems are based on machine learning.
3. **Voice assistants** – Products like Amazon Alexa, Microsoft Cortana, Apple Siri, and Google Assistant are all examples of voice assistants. They take speech commands and perform tasks. These chatbots are the result of machine learning technologies.
4. **Technologies like Google Maps and those used by Uber** are all examples of machine learning which offer to locate and navigate shortest paths to reduce time.

The machine learning applications are enormous. The following Table 1.4 summarizes some of the machine learning applications.

Table 1.4: Applications' Survey Table

S.No.	Problem Domain	Applications
1.	Business	Predicting the bankruptcy of a business firm
2.	Banking	Prediction of bank loan defaulters and detecting credit card frauds
3.	Image Processing	Image search engines, object identification, image classification, and generating synthetic images
4.	Audio/Voice	Chatbots like Alexa, Microsoft Cortana. Developing chatbots for customer support, speech to text, and text to voice
5.	Telecommuni- cation	Trend analysis and identification of bogus calls, fraudulent calls and its callers, churn analysis
6.	Marketing	Retail sales analysis, market basket analysis, product performance analysis, market segmentation analysis, and study of travel patterns of customers for marketing tours
7.	Games	Game programs for Chess, GO, and Atari video games
8.	Natural Language Translation	Google Translate, Text summarization, and sentiment analysis
9.	Web Analysis and Services	Identification of access patterns, detection of e-mail spams, viruses, personalized web services, search engines like Google, detection of promotion of user websites, and finding loyalty of users after web page layout modification
10.	Medicine	Prediction of diseases, given disease symptoms as cancer or diabetes. Prediction of effectiveness of the treatment using patient history and Chatbots to interact with patients like IBM Watson uses machine learning technologies.
11.	Multimedia and Security	Face recognition/identification, biometric projects like identification of a person from a large image or video database, and applications involving multimedia retrieval
12.	Scientific Domain	Discovery of new galaxies, identification of groups of houses based on house type/geographical location, identification of earthquake epicenters, and identification of similar land use

Understanding Data

Machine learning algorithms involve large datasets. Hence, it is necessary to understand the data and datasets before applying machine learning algorithms. This chapter aims to introduce the concepts necessary to understand data better.

Learning Objectives

- Understand data and types of data
- Know about data management
- Learn the basics of dataset
- Identify the data types
- Introduce basics of descriptive analytics
- Explore data visualization techniques
- Introduce univariate data analysis
- Understand basics of bivariate and multivariate data analysis

2.1 WHAT IS DATA?

Scan for information on 'Machine Learning and Importance of Linear Algebra', 'Matrices and Tensors', 'Sampling Techniques', 'Information Theory', 'Evaluation of Classifier Model' and 'Additional Examples'



All facts are data. In computer systems, bits encode facts present in numbers, text, images, audio, and video. Data can be directly human interpretable (such as numbers or texts) or diffused data such as images or video that can be interpreted only by a computer. Today, business organizations are accumulating vast and growing amounts of data of the order of gigabytes, tera bytes, exabytes. A byte is 8 bits. A bit is either 0 or 1. A kilo byte (KB) is 1024 bytes, one mega byte (MB) is approximately 1000 KB, one giga byte is approximately 1,000,000 KB, 1000 giga bytes is one tera byte and 1000000 tera bytes is one Exa byte.

Data is available in different data sources like flat files, databases, or data warehouses. It can either be an operational data or a non-operational data. Operational data is the one that is encountered in normal business procedures and processes. For example, daily sales data is operational data, on the other hand, non-operational data is the kind of data that is used for decision making.

Data by itself is meaningless. It has to be processed to generate any information. A string of bytes is meaningless. Only when a label is attached like height of students of a class, the data becomes meaningful. Processed data is called information that includes patterns, associations, or relationships among data. For example, sales data can be analyzed to extract information like which product was sold larger in the last quarter of the year.

Elements of Big Data

Data whose volume is less and can be stored and processed by a small-scale computer is called 'small data'. These data are collected from several sources, and integrated and processed by a small-scale computer. Big data, on the other hand, is a larger data whose volume is much larger than 'small data' and is characterized as follows:

1. **Volume** – Since there is a reduction in the cost of storing devices, there has been a tremendous growth of data. Small traditional data is measured in terms of gigabytes (GB) and terabytes (TB), but Big Data is measured in terms of petabytes (PB) and exabytes (EB). One exabyte is 1 million terabytes.
2. **Velocity** – The fast arrival speed of data and its increase in data volume is noted as velocity. The availability of IoT devices and Internet power ensures that the data is arriving at a faster rate. Velocity helps to understand the relative growth of big data and its accessibility by users, systems and applications.
3. **Variety** – The variety of Big Data includes:
 - **Form** – There are many forms of data. Data types range from text, graph, audio, video, to maps. There can be composite data too, where one media can have many other sources of data, for example, a video can have an audio song.
 - **Function** – These are data from various sources like human conversations, transaction records, and old archive data.
 - **Source of data** – This is the third aspect of variety. There are many sources of data. Broadly, the data source can be classified as open/public data, social media data and multimodal data. These are discussed in Section 2.3.1 of this chapter.

Some of the other forms of Vs that are often quoted in the literature as characteristics of Big data are:

4. **Veracity of data** – Veracity of data deals with aspects like conformity to the facts, truthfulness, believability, and confidence in data. There may be many sources of error such as technical errors, typographical errors, and human errors. So, veracity is one of the most important aspects of data.
5. **Validity** – Validity is the accuracy of the data for taking decisions or for any other goals that are needed by the given problem.
6. **Value** – Value is the characteristic of big data that indicates the value of the information that is extracted from the data and its influence on the decisions that are taken based on it.

Thus, these 6 Vs are helpful to characterize the big data. The data quality of the numeric attributes is determined by factors like precision, bias, and accuracy. Precision is defined as the closeness of repeated measurements. Often, standard deviation is used to measure the precision. Bias is a systematic result due to erroneous assumptions of the algorithms or procedures. Accuracy is the degree of measurement of errors that refers to the closeness of measurements to the true value of the quantity. Normally, the significant digits used to store and manipulate indicate the accuracy of the measurement.

2.1.1 Types of Data

In Big Data, there are three kinds of data. They are structured data, unstructured data, and semi-structured data.

Structured Data

In structured data, data is stored in an organized manner such as a database where it is available in the form of a table. The data can also be retrieved in an organized manner using tools like SQL.

The structured data frequently encountered in machine learning are listed below:

Record Data A dataset is a collection of measurements taken from a process. We have a collection of objects in a dataset and each object has a set of measurements. The measurements can be arranged in the form of a matrix. Rows in the matrix represent an object and can be called as entities, cases, or records. The columns of the dataset are called attributes, features, or fields. The table is filled with observed data. Also, it is better to note the general jargons that are associated with the dataset. Label is the term that is used to describe the individual observations.

Data Matrix It is a variation of the record type because it consists of numeric attributes. The standard matrix operations can be applied on these data. The data is thought of as points or vectors in the multidimensional space where every attribute is a dimension describing the object.

Graph Data It involves the relationships among objects. For example, a web page can refer to another web page. This can be modeled as a graph. The modes are web pages and the hyperlink is an edge that connects the nodes.

Ordered Data Ordered data objects involve attributes that have an implicit order among them.

The examples of ordered data are:

1. **Temporal data** – It is the data whose attributes are associated with time. For example, the customer purchasing patterns during festival time is sequential data. Time series data is a special type of sequence data where the data is a series of measurements over time.
2. **Sequence data** – It is like sequential data but does not have time stamps. This data involves the sequence of words or letters. For example, DNA data is a sequence of four characters – A T G C.
3. **Spatial data** – It has attributes such as positions or areas. For example, maps are spatial data where the points are related by location.

Unstructured Data

Unstructured data includes video, image, and audio. It also includes textual documents, programs, and blog data. It is estimated that 80% of the data are unstructured data.

Semi-Structured Data

Semi-structured data are partially structured and partially unstructured. These include data like XML/JSON data, RSS feeds, and hierarchical data.

2.1.2 Data Storage and Representation

Once the dataset is assembled, it must be stored in a structure that is suitable for data analysis. The goal of data storage management is to make data available for analysis. There are different approaches to organize and manage data in storage files and systems from flat file to data warehouses. Some of them are listed below:

Flat Files These are the simplest and most commonly available data source. It is also the cheapest way of organizing the data. These flat files are the files where data is stored in plain ASCII or EBCDIC format. Minor changes of data in flat files affect the results of the data mining algorithms. Hence, flat file is suitable only for storing small dataset and not desirable if the dataset becomes larger.

Some of the popular spreadsheet formats are listed below:

- CSV files – CSV stands for comma-separated value files where the values are separated by commas. These are used by spreadsheet and database applications. The first row may have attributes and the rest of the rows represent the data.
- TSV files – TSV stands for Tab separated values files where values are separated by Tab.

Both CSV and TSV files are generic in nature and can be shared. There are many tools like Google Sheets and Microsoft Excel to process these files.

Database System It normally consists of database files and a database management system (DBMS). Database files contain original data and metadata. DBMS aims to manage data and improve operator performance by including various tools like database administrator, query processing, and transaction manager. A relational database consists of sets of tables. The tables have rows and columns. The columns represent the attributes and rows represent tuples. A tuple corresponds to either an object or a relationship between objects. A user can access and manipulate the data in the database using SQL.

Different types of databases are listed below:

1. A transactional database is a collection of transactional records. Each record is a transaction. A transaction may have a time stamp, identifier and a set of items, which may have links to other tables. Normally, transactional databases are created for performing associational analysis that indicates the correlation among the items.
2. Time-series database stores time related information like log files where data is associated with a time stamp. This data represents the sequences of data, which represent values or events obtained over a period (for example, hourly, weekly or yearly) or repeated time span. Observing sales of product continuously may yield a time-series data.

3. Spatial databases contain spatial information in a raster or vector format. Raster formats are either bitmaps or pixel maps. For example, images can be stored as a raster data. On the other hand, the vector format can be used to store maps as maps use basic geometric primitives like points, lines, polygons and so forth.

World Wide Web (WWW) It provides a diverse, worldwide online information source. The objective of data mining algorithms is to mine interesting patterns of information present in WWW.

XML (eXtensible Markup Language) It is both human and machine interpretable data format that can be used to represent data that needs to be shared across the platforms.

Data Stream It is dynamic data, which flows in and out of the observing environment. Typical characteristics of data stream are huge volume of data, dynamic, fixed order movement, and real-time constraints.

RSS (Really Simple Syndication) It is a format for sharing instant feeds across services.

JSON (JavaScript Object Notation) It is another useful data interchange format that is often used for many machine learning algorithms.

2.2 BIG DATA ANALYTICS AND TYPES OF ANALYTICS

The primary aim of data analysis is to assist business organizations to take decisions. For example, a business organization may want to know which is the fastest selling product, in order for them to market activities. Data analysis is an activity that takes the data and generates useful information and insights for assisting the organizations.

Data analysis and data analytics are terms that are used interchangeably to refer to the same concept. However, there is a subtle difference. Data analytics is a general term and data analysis is a part of it. Data analytics refers to the process of data collection, preprocessing and analysis. It deals with the complete cycle of data management. Data analysis is just analysis and is a part of data analytics. It takes historical data and does the analysis. Data analytics, instead, concentrates more on future and helps in prediction.

There are four types of data analytics:

1. Descriptive analytics
2. Diagnostic analytics
3. Predictive analytics
4. Prescriptive analytics

Descriptive Analytics It is about describing the main features of the data. After data collection is done, descriptive analytics deals with the collected data and quantifies it. It is often stated that analytics is essentially statistics. There are two aspects of statistics – Descriptive and Inference. Descriptive analytics only focuses on the description part of the data and not the inference part.

Diagnostic Analytics It deals with the question – ‘Why?’. This is also known as causal analysis, as it aims to find out the cause and effect of the events. For example, if a product is not selling, diagnostic analytics aims to find out the reason. There may be multiple reasons and associated effects are analyzed as part of it.

Predictive Analytics It deals with the future. It deals with the question – ‘What will happen in future given this data?’. This involves the application of algorithms to identify the patterns to predict the future. The entire course of machine learning is mostly about predictive analytics and forms the core of this book.

Prescriptive Analytics It is about the finding the best course of action for the business organizations. Prescriptive analytics goes beyond prediction and helps in decision making by giving a set of actions. It helps the organizations to plan better for the future and to mitigate the risks that are involved.

2.3 BIG DATA ANALYSIS FRAMEWORK

For performing data analytics, many frameworks are proposed. All proposed analytics frameworks have some common factors. Big data framework is a layered architecture. Such an architecture has many advantages such as genericness. A 4-layer architecture has the following layers:

1. Data connection layer
2. Data management layer
3. Data analytics later
4. Presentation layer

Data Connection Layer It has data ingestion mechanisms and data connectors. Data ingestion means taking raw data and importing it into appropriate data structures. It performs the tasks of ETL process. By ETL, it means extract, transform and load operations.

Data Management Layer It performs preprocessing of data. The purpose of this layer is to allow parallel execution of queries, and read, write and data management tasks. There may be many schemes that can be implemented by this layer such as data-in-place, where the data is not moved at all, or constructing data repositories such as data warehouses and pull data on-demand mechanisms.

Data Analytic Layer It has many functionalities such as statistical tests, machine learning algorithms to understand, and construction of machine learning models. This layer implements many model validation mechanisms too. The processing is done as shown in Box 2.1.

Box 2.1: Types of Processing

Cloud Computing

Cloud computing is an emerging technology which is basically a business service model or simply called as pay-per-usage model. The term ‘Cloud’ refers to the Internet that provides sharing of processing power, applications, storage and services. It offers different kinds of services such as Iaas, Paas, and SaaS.

SaaS (Software as a Service) enables users to access software applications from the cloud. Paas (Platform as a Service) provides users the platform to develop and run their applications. Iaas (Infrastructure as a Service) enables users to access the infrastructure required to run their applications, storage, operating systems, etc.

(Continued)

The cloud services can be deployed in four most commonly used deployment models such as Public Cloud, Private Cloud, Community Cloud, and Hybrid Cloud based on the service model, organization, geographic location, etc. The **Public Cloud** is accessible to the public and is owned by a vendor, who offers the services of the cloud to the users. **Private Cloud** is a privately-owned cloud where the user or an organization owns the cloud and only the user or employees of that organization have access to the cloud, thereby making data and transactions secure. In **Community Cloud**, the infrastructure is owned jointly by different organizations. The **Hybrid Cloud** is the combination of two or more cloud types. The characteristics of cloud computing are:

1. Shared Infrastructure – Sharing of physical services, storage, and networking capabilities
2. Dynamic Provisioning – Resources assigned dynamically, based on demands
3. Dynamic Scaling – Expansion and contraction of service capability
4. Network Access – Needs to be accessed across the internet
5. Utility-based Metering – Uses metering to provide reporting and billing information
6. Multitenancy – Serves multiple customers
7. Reliability – Customer reliable service

Grid Computing

Grid Computing is a parallel and distributed computing framework consisting of a network of computers offering a super computing service as a single virtual supercomputer. This high-performance computing is required to perform specialized tasks that require a high computing power and a single computer cannot provide enough computing resources. The grid computing model forms a grid by connecting tens of thousands of nodes as a cluster that runs on an operating system. In this model, the resources are pooled together and the load is shared across multiple nodes to accomplish a task more quickly. This grid is constructed by middleware software that evenly distributes the task to several nodes connected in the grid. The individual nodes perform the task independently and in parallel which are then integrated to complete the large-scale task. This model of computing is best suited for applications that are complex and can be computed in parallel.

H-Computing (High Performance Computing or HPC)

It enables to perform complex tasks at high speed. It aggregates computing power in such a way that provides much higher performance to solve complex problems in science, engineering, research or business. It leverages parallel processing techniques for solving complex computational problems. HPC system achieves this sustained performance through concurrent use of computing resources. An HPC system combines the computing power of thousands of compute nodes that work in parallel to complete tasks faster. The system comprises three key components called compute, network and storage. The architecture of HPC consists of compute servers that are networked together to form a cluster. Software programs are run in parallel on the servers in the cluster and are networked to the data storage to capture the output. These components work together to complete a task.

Presentation Layer It has mechanisms such as dashboards, and applications that display the results of analytical engines and machine learning algorithms.

Thus, the Big Data processing cycle involves data management that consists of the following steps.

1. Data collection
2. Data preprocessing
3. Applications of machine learning algorithm
4. Interpretation of results and visualization of machine learning algorithm

This is an iterative process and is carried out on a permanent basis to ensure that data is suitable for data mining.

Application and interpretation of machine learning algorithms constitute the basis for the rest of the book. So, primarily, data collection and data preprocessing are covered as part of this chapter.

The following section covers data collection in detail.

2.3.1 Data Collection

The first task of gathering datasets are the collection of data. It is often estimated that most of the time is spent for collection of good quality data. A good quality data yields a better result. It is often difficult to characterize a 'Good data'. 'Good data' is one that has the following properties:

1. Timeliness – The data should be relevant and not stale or obsolete data.
2. Relevancy – The data should be relevant and ready for the machine learning or data mining algorithms. All the necessary information should be available and there should be no bias in the data.
3. Knowledge about the data – The data should be understandable and interpretable, and should be self-sufficient for the required application as desired by the domain knowledge engineer.

Broadly, the data source can be classified as open/public data, social media data and multimodal data.

1. Open or public data source – It is a data source that does not have any stringent copyright rules or restrictions. Its data can be primarily used for many purposes. Government census data are good examples of open data:
 - Digital libraries that have huge amount of text data as well as document images
 - Scientific domains with a huge collection of experimental data like genomic data and biological data
 - Healthcare systems that use extensive databases like patient databases, health insurance data, doctors' information, and bioinformatics information
2. Social media – It is the data that is generated by various social media platforms like Twitter, Facebook, YouTube, and Instagram. An enormous amount of data is generated by these platforms.
3. Multimodal data – It includes data that involves many modes such as text, video, audio and mixed types. Some of them are listed below:

- Image archives contain larger image databases along with numeric and text data
 - The World Wide Web (WWW) has huge amount of data that is distributed on the Internet.
- These data are heterogeneous in nature.

2.3.2 Data Preprocessing

In real world, the available data is 'dirty'. By this word 'dirty', it means:

- Incomplete data
- Inaccurate data
- Outlier data
- Data with missing values
- Data with inconsistent values
- Duplicate data

Data preprocessing improves the quality of the data mining techniques. The raw data must be preprocessed to give accurate results. The process of detection and removal of errors in data is called data cleaning. Data wrangling means making the data processable for machine learning algorithms. Some of the data errors include human errors such as typographical errors or incorrect measurement and structural errors like improper data formats. Data errors can also arise from omission and duplication of attributes. Noise is a random component and involves distortion of a value or introduction of spurious objects. Often, the noise is used if the data is a spatial or temporal component. Certain deterministic distortions in the form of a streak are known as artifacts.

Consider, for example, the following patient Table 2.1. The 'bad' or 'dirty' data can be observed in this table.

Table 2.1: Illustration of 'Bad' Data

Patient ID	Name	Age	Date of Birth (DoB)	Fever	Salary
1.	John	21		Low	-1500
2.	Andre	36		High	Yes
3.	David	5	10/10/1980	Low	" "
4.	Raju	136		High	Yes

It can be observed that data like Salary = ' ' is incomplete data. The DoB of patients, John, Andre, and Raju, is the missing data. The age of David is recorded as '5' but his DoB indicates it is 10/10/1980. This is called inconsistent data.

Inconsistent data occurs due to problems in conversions, inconsistent formats, and difference in units. Salary for John is -1500. It cannot be less than '0'. It is an instance of noisy data. Outliers are data that exhibit the characteristics that are different from other data and have very unusual values. The age of Raju cannot be 136. It might be a typographical error. It is often required to distinguish between noise and outlier data.

Outliers may be legitimate data and sometimes are of interest to the data mining algorithms. These errors often come during data collection stage. These must be removed so that machine learning algorithms yield better results as the quality of results is determined by the quality of input data. This removal process is called data cleaning.

Missing Data Analysis

The primary data cleaning process is missing data analysis. Data cleaning routines attempt to fill up the missing values, smoothen the noise while identifying the outliers and correct the inconsistencies of the data. This enables data mining to avoid overfitting of the models.

The procedures that are given below can solve the problem of missing data:

1. Ignore the tuple – A tuple with missing data, especially the class label, is ignored. This method is not effective when the percentage of the missing values increases.
2. Fill in the values manually – Here, the domain expert can analyse the data tables and carry out the analysis and fill in the values manually. But, this is time consuming and may not be feasible for larger sets.
3. A global constant can be used to fill in the missing attributes. The missing values may be 'Unknown' or be 'Infinity'. But, some data mining results may give spurious results by analysing these labels.
4. The attribute value may be filled by the attribute value. Say, the average income can replace a missing value.
5. Use the attribute mean for all samples belonging to the same class. Here, the average value replaces the missing values of all tuples that fall in this group.
6. Use the most possible value to fill in the missing value. The most probable value can be obtained from other methods like classification and decision tree prediction.

Some of these methods introduce bias in the data. The filled value may not be correct and could be just an estimated value. Hence, the difference between the estimated and the original value is called an error or bias.

Removal of Noisy or Outlier Data

Noise is a random error or variance in a measured value. It can be removed by using binning, which is a method where the given data values are sorted and distributed into equal frequency bins. The bins are also called as buckets. The binning method then uses the neighbor values to smooth the noisy data.

Some of the techniques commonly used are 'smoothing by means' where the mean of the bin removes the values of the bins, 'smoothing by bin medians' where the bin median replaces the bin values, and 'smoothing by bin boundaries' where the bin value is replaced by the closest bin boundary. The maximum and minimum values are called bin boundaries. Binning methods may be used as a discretization technique. Example 2.1 illustrates this principle.

Example 2.1: Consider the following set: $S = \{12, 14, 19, 22, 24, 26, 28, 31, 34\}$. Apply various binning techniques and show the result.

Solution: By equal-frequency bin method, the data should be distributed across bins. Let us assume the bins of size 3, then the above data is distributed across the bins as shown below:

Bin 1 : 12, 14, 19

Bin 2 : 22, 24, 26

Bin 3 : 28, 31, 32

By smoothing bins method, the bins are replaced by the bin means. This method results in:

Bin 1 : 15, 15, 15

Bin 2 : 24, 24, 24

Bin 3 : 30.3, 30.3, 30.3

Using smoothing by bin boundaries method, the bins' values would be like:

Bin 1 : 12, 12, 19

Bin 2 : 22, 22, 26

Bin 3 : 28, 32, 32

As per the method, the minimum and maximum values of the bin are determined, and it serves as bin boundary and does not change. Rest of the values are transformed to the nearest value. It can be observed in Bin 1, the middle value 14 is compared with the boundary values 12 and 19 and changed to the closest value, that is 12. This process is repeated for all bins.

Data Integration and Data Transformations

Data integration involves routines that merge data from multiple sources into a single data source. So, this may lead to redundant data. The main goal of data integration is to detect and remove redundancies that arise from integration. Data transformation routines perform operations like normalization to improve the performance of the data mining algorithms. It is necessary to transform data so that it can be processed. This can be considered as a preliminary stage of data conditioning. Normalization is one such technique. In normalization, the attribute values are scaled to fit in a range (say 0–1) to improve the performance of the data mining algorithm. Often, in neural networks, these techniques are used. Some of the normalization procedures used are:

1. *Min-Max*
2. *z-Score*

Min-Max Procedure It is a normalization technique where each variable V is normalized by its difference with the minimum value divided by the range to a new range, say 0–1. Often, neural networks require this kind of normalization. The formula to implement this normalization is given as:

$$\text{min-max} = \frac{V - \text{min}}{\text{max} - \text{min}} \times (\text{new max} - \text{new min}) + \text{new min} \quad (2.1)$$

Here max-min is the range. Min and max are the minimum and maximum of the given data, new max and new min are the minimum and maximum of the target range, say 0 and 1.

Example 2.2: Consider the set: $V = \{88, 90, 92, 94\}$. Apply *Min-Max* procedure and map the marks to a new range 0–1.

Solution: The minimum of the list V is 88 and maximum is 94. The new min and new max are 0 and 1, respectively. The mapping can be done using Eq. (2.1) as:

For marks 88,

$$\text{min-max} = \frac{88 - 88}{94 - 88} \times (1 - 0) + 0 = 0$$

Similarly, other marks can be computed as follows:

For marks 90,

$$\text{min-max} = \frac{90 - 88}{94 - 88} \times (1 - 0) + 0 = 0.33$$

For marks 92,

$$\text{min-max} = \frac{92 - 88}{94 - 88} \times (1 - 0) + 0 = \frac{4}{6} = 0.66$$

For marks 94,

$$\text{min-max} = \frac{94 - 88}{94 - 88} \times (1 - 0) + 0 = \frac{6}{6} = 1$$

So, it can be observed that the marks {88, 90, 92, 94} are mapped to the new range [0, 0.33, 0.66, 1].

Thus, the *Min-Max* normalization range is between 0 and 1.

z-Score Normalization This procedure works by taking the difference between the field value and mean value, and by scaling this difference by standard deviation of the attribute.

$$V^* = V - \mu/\sigma \quad (2.2)$$

Here, σ is the standard deviation of the list V and μ is the mean of the list V .

Example 2.3: Consider the mark list $V = \{10, 20, 30\}$, convert the marks to z-score.

Solution: The mean and Sample Standard deviation (σ) values of the list V are 20 and 10, respectively. So the z-scores of these marks are calculated using Eq. (2.2) as:

$$\text{z-score of } 10 = \frac{10 - 20}{10} = -\frac{10}{10} = -1$$

$$\text{z-score of } 20 = \frac{20 - 20}{10} = \frac{0}{10} = 0$$

$$\text{z-score of } 30 = \frac{30 - 20}{10} = \frac{10}{10} = 1$$

Hence, the z-score of the marks 10, 20, 30 are -1, 0 and 1, respectively.

What is the use of z-scores? z-scores are used to detect outlier detection. If the data value z-score function is either less than -3 or greater than +3, then it is possibly an outlier. The major disadvantage of z-score function is that it is extremely sensitive to outliers as it is dependent on mean.

Data Reduction

Data reduction reduces data size but produces the same results. There are different ways in which data reduction can be carried out such as data aggregation, feature selection, and dimensionality reduction.

2.4 DESCRIPTIVE STATISTICS

Descriptive statistics is a branch of statistics that does dataset summarization. It is used to summarize and describe data. Descriptive statistics are just descriptive and do not go beyond that. In other words, descriptive statistics do not bother too much about machine learning algorithms and its functioning.

Data visualization is a branch of study that is useful for investigating the given data. Mainly, the plots are useful to explain and present data to customers.

Descriptive analytics and data visualization techniques help to understand the nature of the data, which further helps to determine the kinds of machine learning or data mining tasks that can be applied to the data. This step is often known as Exploratory Data Analysis (EDA). The focus of EDA is to understand the given data and to prepare it for machine learning algorithms. EDA includes descriptive statistics and data visualization.

Let us discuss descriptive statistics with the fundamental concepts of datatypes.

Dataset and Data Types

A dataset can be assumed to be a collection of data objects. The data objects may be records, points, vectors, patterns, events, cases, samples or observations. These records contain many attributes. An attribute can be defined as the property or characteristics of an object.

For example, consider the following database shown in sample Table 2.2.

Table 2.2: Sample Patient Table

Patient ID	Name	Age	Blood Test	Fever	Disease
1.	John	21	Negative	Low	No
2.	Andre	36	Positive	High	Yes

Every attribute should be associated with a value. This process is called measurement. The type of attribute determines the data types, often referred to as measurement scale types. The data types are shown in Figure 2.1.

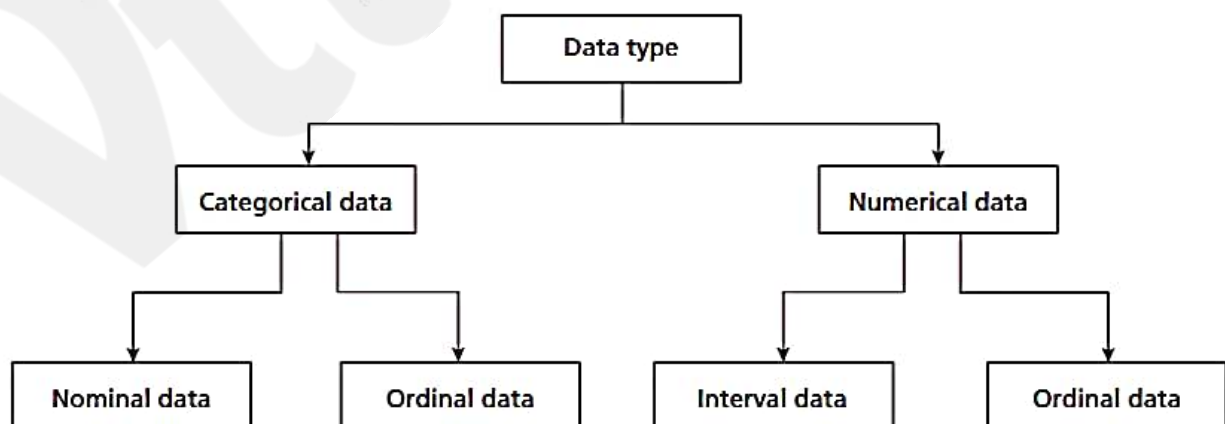


Figure 2.1: Types of Data

Broadly, data can be classified into two types:

1. Categorical or qualitative data
2. Numerical or quantitative data

Categorical or Qualitative Data The categorical data can be divided into two types. They are nominal type and ordinal type.

- **Nominal Data** – In Table 2.2, patient ID is nominal data. Nominal data are symbols and cannot be processed like a number. For example, the average of a patient ID does not make any statistical sense. Nominal data type provides only information but has no ordering among data. Only operations like ($=$, $\neq$) are meaningful for these data. For example, the patient ID can be checked for equality and nothing else.
- **Ordinal Data** – It provides enough information and has natural order. For example, Fever = {Low, Medium, High} is an ordinal data. Certainly, low is less than medium and medium is less than high, irrespective of the value. Any transformation can be applied to these data to get a new value.

Numeric or Quantitative Data It can be divided into two categories. They are interval type and ratio type.

- **Interval Data** – Interval data is a numeric data for which the differences between values are meaningful. For example, there is a difference between 30 degree and 40 degree. Only the permissible operations are $+$ and $-$.
- **Ratio Data** – For ratio data, both differences and ratio are meaningful. The difference between the ratio and interval data is the position of zero in the scale. For example, take the Centigrade-Fahrenheit conversion. The zeroes of both scales do not match. Hence, these are interval data.

Another way of classifying the data is to classify it as:

1. Discrete value data
2. Continuous data

Discrete Data This kind of data is recorded as integers. For example, the responses of the survey can be discrete data. Employee identification number such as 10001 is discrete data.

Continuous Data It can be fitted into a range and includes decimal point. For example, age is a continuous data. Though age appears to be discrete data, one may be 12.5 years old and it makes sense. Patient height and weight are all continuous data.

Third way of classifying the data is based on the number of variables used in the dataset. Based on that, the data can be classified as univariate data, bivariate data, and multivariate data. This is shown in Figure 2.2.

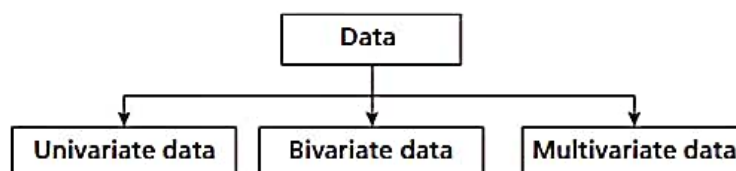


Figure 2.2: Types of Data Based on Variables

In case of univariate data, the dataset has only one variable. A variable is also called as category. Bivariate data indicates that the number of variables used are two and multivariate data uses three or more variables.

This chapter primarily deals with univariate data in detail with just an overview of bivariate and multivariate data.

2.5 UNIVARIATE DATA ANALYSIS AND VISUALIZATION

Scan for information on 'Measures of Frequency' and also for 'Additional Examples'



Univariate analysis is the simplest form of statistical analysis. As the name indicates, the dataset has only one variable. A variable can be called as a category. Univariate does not deal with cause or relationships. The aim of univariate analysis is to describe data and find patterns.

Univariate data description involves finding the frequency distributions, central tendency measures, dispersion or variation, and shape of the data.

2.5.1 Data Visualization

To understand data, graph visualization is must. Data visualization helps to understand data. It helps to present information and data to customers. Some of the graphs that are used in univariate data analysis are bar charts, histograms, frequency polygons and pie charts.

The advantages of the graphs are presentation of data, summarization of data, description of data, exploration of data, and to make comparisons of data. Let us consider some forms of graphs now:

Bar Chart A Bar chart (or Bar graph) is used to display the frequency distribution for variables. Bar charts are used to illustrate discrete data. The charts can also help to explain the counts of nominal data. It also helps in comparing the frequency of different groups.

The bar chart for students' marks {45, 60, 60, 80, 85} with Student ID = {1, 2, 3, 4, 5} is shown below in Figure 2.3.

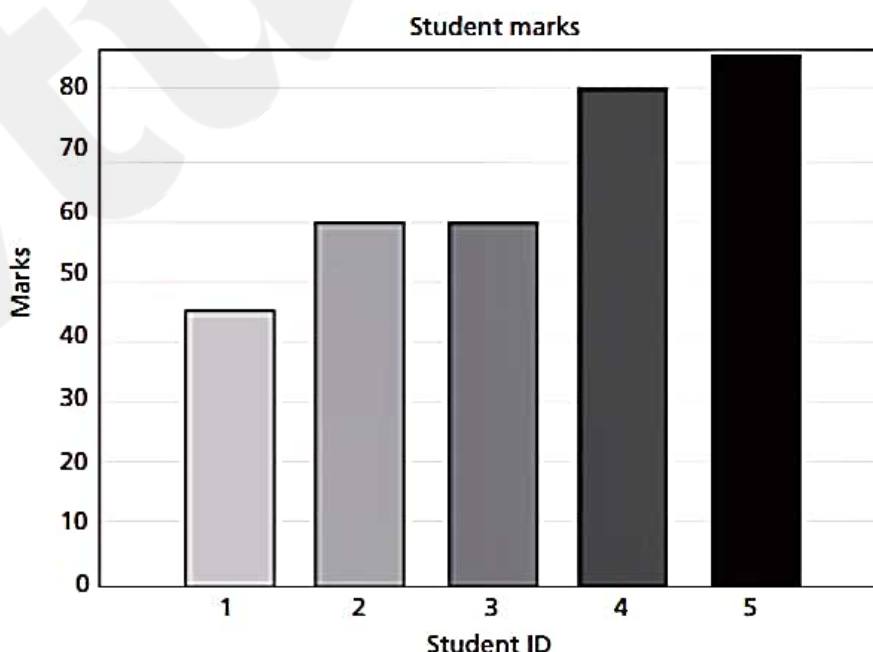


Figure 2.3: Bar Chart

Pie Chart These are equally helpful in illustrating the univariate data. The percentage frequency distribution of students' marks {22, 22, 40, 40, 70, 70, 70, 85, 90, 90} is below in Figure 2.4.

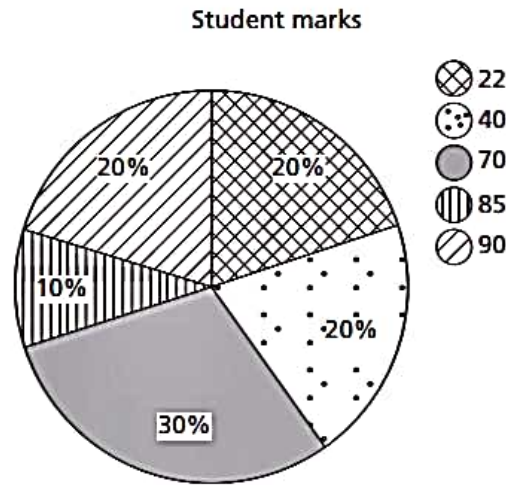


Figure 2.4: Pie Chart

It can be observed that the number of students with 22 marks are 2. The total number of students are 10. So, $2/10 \times 100 = 20\%$ space in a pie of 100% is allotted for marks 22 in Figure 2.4.

Histogram It plays an important role in data mining for showing frequency distributions. The histogram for students' marks {45, 60, 60, 80, 85} in the group range of 0–25, 26–50, 51–75, 76–100 is given below in Figure 2.5. One can visually inspect from Figure 2.5 that the number of students in the range 76–100 is 2.

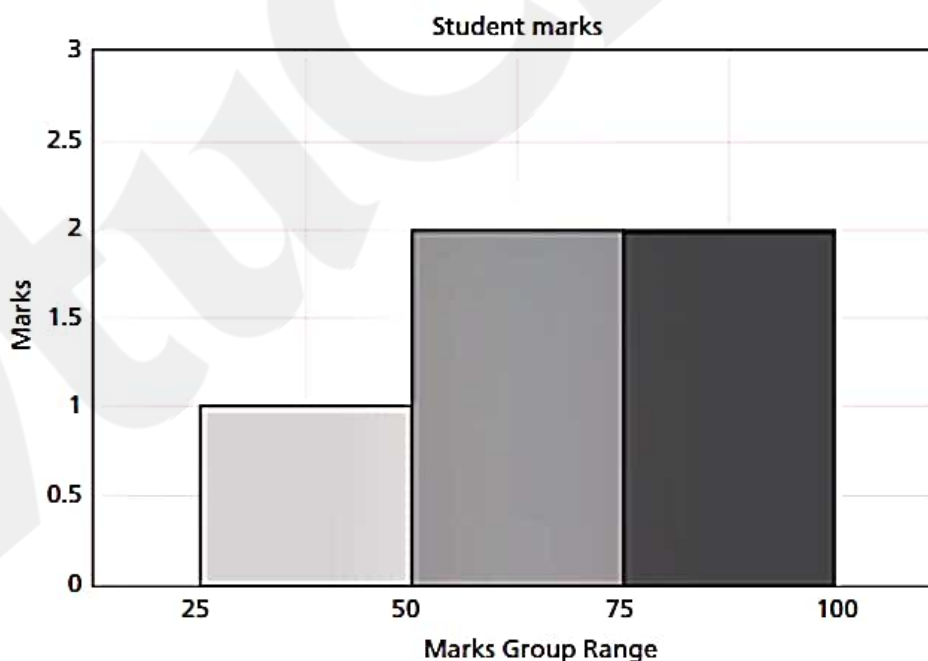


Figure 2.5: Sample Histogram of English Marks

Histogram conveys useful information like nature of data and its mode. Mode indicates the peak of dataset. In other words, histograms can be used as charts to show frequency, skewness present in the data, and shape.

Dot Plots These are similar to bar charts. They are less clustered as compared to bar charts, as they illustrate the bars only with single points. The dot plot of English marks for five students with ID as {1, 2, 3, 4, 5} and marks {45, 60, 60, 80, 85} is given in Figure 2.6. The advantage is that by visual inspection one can find out who got more marks.

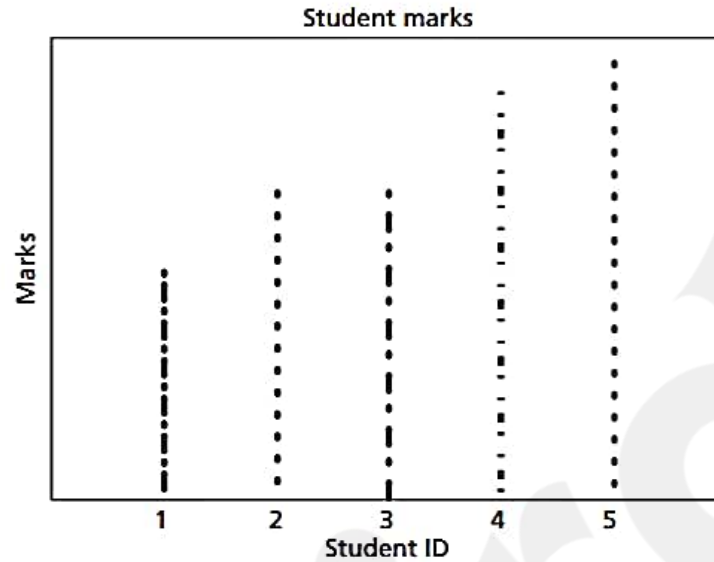


Figure 2.6: Dot Plots

2.5.2 Central Tendency

One cannot remember all the data. Therefore, a condensation or summary of the data is necessary. This makes the data analysis easy and simple. One such summary is called central tendency. Thus, central tendency can explain the characteristics of data and that further helps in comparison. Mass data have tendency to concentrate at certain values, normally in the central location. It is called measure of central tendency (or averages). This represents the first order of measures. Popular measures are mean, median and mode.

1. **Mean – Arithmetic average (or mean)** is a measure of central tendency that represents the 'center' of the dataset. This is the commonest measure used in our daily conversation such as average income or average traffic. It can be found by adding all the data and dividing the sum by the number of observations. Mathematically, the *average* of all the values in the sample (population) is denoted as $\bar{x}$. Let $x_1, x_2, \dots, x_N$ be a set of 'N' values or observations, then the arithmetic mean is given as:

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_N}{N} = \frac{1}{N} \sum_{i=1}^N x_i \quad (2.3)$$

For example, the mean of the three numbers 10, 20, and 30 is $\frac{10 + 20 + 30}{3} = \frac{60}{3} = 20$

- **Weighted mean** – Unlike arithmetic mean that gives the weightage of all items equally, weighted mean gives different importance to all items as the item importance varies. Hence, different weightage can be given to items.

In case of frequency distribution, mid values of the range are taken for computation. This is illustrated in the following computation.

In weighted mean, the mean is computed by adding the product of proportion and group mean. It is mostly used when the sample sizes are unequal.

- **Geometric mean** – Let $x_1, x_2, \dots, x_N$ be a set of 'N' values or observations. Geometric mean is the N^{th} root of the product of N items. The formula for computing geometric mean is given as follows:

$$\text{Geometric mean} = \left(\prod_{i=1}^n x_i \right)^{\frac{1}{N}} = \sqrt[N]{x_1 \times x_2 \times \dots \times x_N} \quad (2.4)$$

Here, n is the number of items and x_i are values. For example, if the values are 6 and 8, the geometric mean is given as $\sqrt[2]{6 \times 8} = \sqrt{48}$. In larger cases, computing geometric mean is difficult. Hence, it is usually calculated as:

$$\text{Anti-log of } \frac{\log(x_1) + \log(x_2) + \dots + \log(x_N)}{N} \quad (2.5)$$

$$= \text{anti-log } \frac{\sum_{i=1}^n \log(x_i)}{N} \quad (2.6)$$

The problem of mean is its extreme sensitiveness to noise. Even small changes in the input affect the mean drastically. Hence, often the top 2% is chopped off and then the mean is calculated for a larger dataset.

2. **Median** – The middle value in the distribution is called median. If the total number of items in the distribution is odd, then the middle value is called median. If the numbers are even, then the average value of two items in the centre is the median. It can be observed that the median is the value where x_i is divided into two equal halves, with half of the values being lower than the median and half higher than the median. A median class is that class where $(N/2)^{\text{th}}$ item is present.

In the continuous case, the median is given by the formula:

$$\text{Median} = L_1 + \frac{\frac{N}{2} - cf}{f} \times i \quad (2.7)$$

Median class is that class where $N/2^{\text{th}}$ item is present. Here, i is the class interval of the median class and L_1 is the lower limit of median class, f is the frequency of the median class, and cf is the cumulative frequency of all classes preceding median.

3. **Mode** – Mode is the value that occurs more frequently in the dataset. In other words, the value that has the highest frequency is called mode. Mode is only for discrete data and is not applicable for continuous data as there are no repeated values in continuous data.

The procedure for finding the mode is to calculate the frequencies for all the values in the data, and mode is the value (or values) with the highest frequency. Normally, the dataset is classified as unimodal, bimodal and trimodal with modes 1, 2 and 3, respectively.

2.5.3 Dispersion

The spreadout of a set of data around the central tendency (mean, median or mode) is called dispersion. Dispersion is represented by various ways such as range, variance, standard deviation, and standard error. These are second order measures. The most common measures of the dispersion data are listed below:

Range Range is the difference between the maximum and minimum of values of the given list of data.

Standard Deviation The mean does not convey much more than a middle point. For example, the following datasets {10, 20, 30} and {10, 50, 0} both have a mean of 20. The difference between these two sets is the spread of data.

Standard deviation is the average distance from the mean of the dataset to each point. The formula for sample standard deviation is given by:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1}} \quad (2.8)$$

Here, N is the size of the population, x_i is observation or value from the population and μ is the population mean. Often, $N - 1$ is used instead of N in the denominator of Eq. (2.8). The reason is that for larger real-world, the division by $N - 1$ gives an answer closer to the actual value.

Quartiles and Inter Quartile Range It is sometimes convenient to subdivide the dataset using coordinates. Percentiles are about data that are less than the coordinates by some percentage of the total value. k^{th} percentile is the property that the $k\%$ of the data lies at or below X_r . For example, median is 50th percentile and can be denoted as $Q_{0.50}$. The 25th percentile is called first quartile (Q_1) and the 75th percentile is called third quartile (Q_3).

Another measure that is useful to measure dispersion is Inter Quartile Range (IQR). The IQR is the difference between Q_3 and Q_1 .

$$\text{Interquartile percentile} = Q_3 - Q_1 \quad (2.9)$$

Outliers are normally the values falling apart at least by the amount $1.5 \times \text{IQR}$ above the third quartile or below the first quartile.

$$\text{Interquartile is defined by } Q_{0.75} - Q_{0.25} \quad (2.10)$$

Example 2.4: For patients' age list {12, 14, 19, 22, 24, 26, 28, 31, 34}, find the IQR.

Solution: The median is in the fifth position. In this case, 24 is the median. The first quartile is median of the scores below the mean i.e., {12, 14, 19, 22}. Hence, it's the median of the list below 24. In this case, the median is the average of the second and third values, that is, $Q_{0.25} = 16.5$. Similarly, the third quartile is the median of the values above the median, that is {26, 28, 31, 34}. So, $Q_{0.75}$ is the average of the seventh and eighth score. In this case, it is $28 + 31/2 = 59/2 = 29.5$.

Hence, the IQR using Eq. (2.10) is:

$$\begin{aligned} &= Q_{0.75} - Q_{0.25} \\ &= 29.5 - 16.5 = 13 \end{aligned}$$

The half of IQR is called semi-quartile range. The Semi Inter Quartile Range (SIQR) is given as:

$$\begin{aligned}\text{SIQR} &= \frac{1}{2} \times \text{IQR} \\ &= \frac{1}{2} \times 13 = 6.5\end{aligned}\quad (2.11)$$

Five-point Summary and Box Plots The median, quartiles Q_1 and Q_3 , and minimum and maximum written in the order $\langle \text{Minimum}, Q_1, \text{Median}, Q_3, \text{Maximum} \rangle$ is known as five-point summary.

Box plots are suitable for continuous variables and a nominal variable. Box plots can be used to illustrate data distributions and summary of data. It is the popular way for plotting five number summaries. A Box plot is also known as a Box and whisker plot.

The box contains bulk of the data. These data are between first and third quartiles. The line inside the box indicates location – mostly median of the data. If the median is not equidistant, then the data is skewed. The whiskers that project from the ends of the box indicate the spread of the tails and the maximum and minimum of the data value.

Example 2.5: Find the 5-point summary of the list {13, 11, 2, 3, 4, 8, 9}.

Solution: The minimum is 2 and the maximum is 13. The Q_1 , Q_2 and Q_3 are 3, 8 and 11, respectively. Hence, 5-point summary is {2, 3, 8, 11, 13}, that is, {minimum, Q_1 , median, Q_3 , maximum}.

Box plots are useful for describing 5-point summary. The Box plot for the set is given in Figure 2.7.

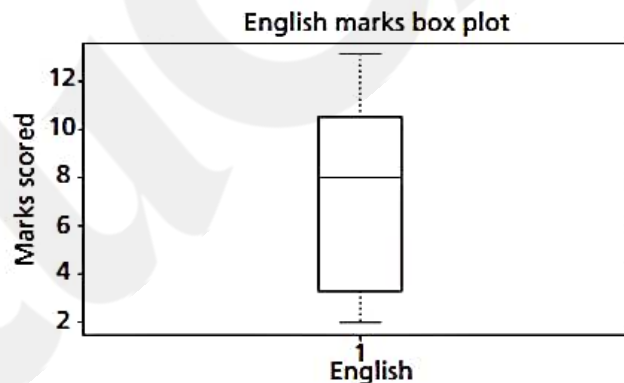


Figure 2.7: Box Plot for English Marks

2.5.4 Shape

Skewness and Kurtosis (called moments) indicate the symmetry/asymmetry and peak location of the dataset.

Skewness

The measures of direction and degree of symmetry are called measures of third order. Ideally, skewness should be zero as in ideal normal distribution. More often, the given dataset may not have perfect symmetry (consider the following Figure 2.8).

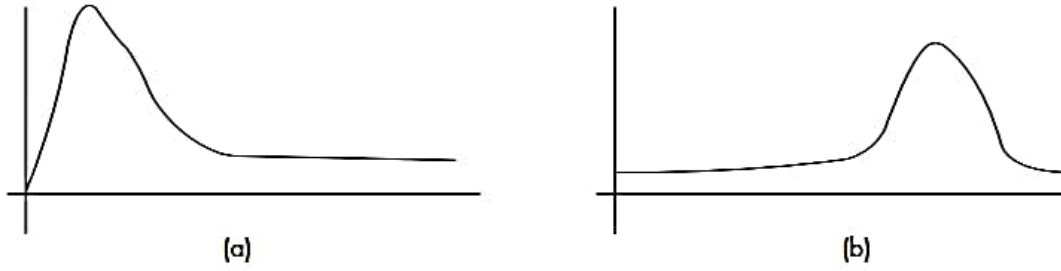


Figure 2.8: (a) Positive Skewed and (b) Negative Skewed Data

The dataset may also either have very high values or extremely low values. If the dataset has far higher values, then it is said to be skewed to the right. On the other hand, if the dataset has far more low values then it is said to be skewed towards left. If the tail is longer on the left-hand side and hump on the right-hand side, it is called positive skew. Otherwise, it is called negative skew.

The given dataset may have an equal distribution of data. The implication of this is that if the data is skewed, then there is a greater chance of outliers in the dataset. This affects the mean and median. Hence, this may affect the performance of the data mining algorithm. A perfect symmetry means the skewness is zero. In the case of skew, the median is greater than the mean. In positive skew, the mean is greater than the median.

Generally, for negatively skewed distribution, the median is more than the mean. The relationship between skew and the relative size of the mean and median can be summarized by a convenient numerical skew index known as Pearson 2 skewness coefficient.

$$\frac{3 \times (\mu - \text{median})}{\sigma} \quad (2.12)$$

Also, the following measure is more commonly used to measure skewness. Let $X_1, X_2, \dots, X_N$ be a set of ' N ' values or observations then the skewness can be given as:

$$\frac{1}{N} \times \sum_{i=1}^N \frac{(x_i - \mu)^3}{\sigma^3} \quad (2.13)$$

Here, μ is the population mean and σ is the population standard deviation of the univariate data. Sometimes, for bias correction instead of N , $N - 1$ is used.

Kurtosis

Kurtosis also indicates the peaks of data. If the data is high peak, then it indicates higher kurtosis and vice versa.

Kurtosis is the measure of whether the data is heavy tailed or light tailed relative to normal distribution. It can be observed that normal distribution has bell-shaped curve with no long tails. Low kurtosis tends to have light tails. The implication is that there is no outlier data. Let $x_1, x_2, \dots, x_N$ be a set of ' N ' values or observations. Then, kurtosis is measured using the formula given below:

$$\frac{\sum_{i=1}^N (x_i - \bar{x})^4 / N}{\sigma^4} \quad (2.14)$$

It can be observed that $N - 1$ is used instead of N in the numerator of Eq. (2.14) for bias correction. Here, $\bar{x}$ and σ are the mean and standard deviation of the univariate data, respectively.

Some of the other useful measures for finding the shape of the univariate dataset are mean absolute deviation (MAD) and coefficient of variation (CV).

Mean Absolute Deviation (MAD)

MAD is another dispersion measure and is robust to outliers. Normally, the outlier point is detected by computing the deviation from median and by dividing it by MAD. Here, the absolute deviation between the data and mean is taken. Thus, the absolute deviation is given as:

$$|x - \mu| \quad (2.15)$$

The sum of the absolute deviations is given as $\sum |x - \mu|$

Therefore, the mean absolute deviation is given as:
$$\frac{\sum |x - \mu|}{N} \quad (2.16)$$

Coefficient of Variation (CV)

Coefficient of variation is used to compare datasets with different units. CV is the ratio of standard deviation and mean, and %CV is the percentage of coefficient of variations.

2.5.5 Special Univariate Plots

The ideal way to check the shape of the dataset is a stem and leaf plot. A stem and leaf plot are a display that help us to know the shape and distribution of the data. In this method, each value is split into a 'stem' and a 'leaf'. The last digit is usually the leaf and digits to the left of the leaf mostly form the stem. For example, marks 45 are divided into stem 4 and leaf 5 in Figure 2.9.

The stem and leaf plot for the English subject marks, say, {45, 60, 60, 80, 85} is given in Figure 2.9.

Stem	Leaf
4	5
5	
6	0 0
7	
8	0 5

Figure 2.9: Stem and Leaf Plot for English Marks

It can be seen from Figure 2.9 that the first column is stem and the second column is leaf. For the given English marks, two students with 60 marks are shown in stem and leaf plot as stem-6 with 2 leaves with 0.

As discussed earlier, the ideal shape of the dataset is a bell-shaped curve. This corresponds to normality. Most of the statistical tests are designed only for normal distribution of data. A Q-Q plot can be used to assess the shape of the dataset. The Q-Q plot is a 2D scatter plot of an univariate data against theoretical normal distribution data or of two datasets – the quartiles of the first and second datasets. The normal Q-Q plot for marks $x = [13 \ 11 \ 2 \ 3 \ 4 \ 8 \ 9]$ is given below in Figure 2.10.

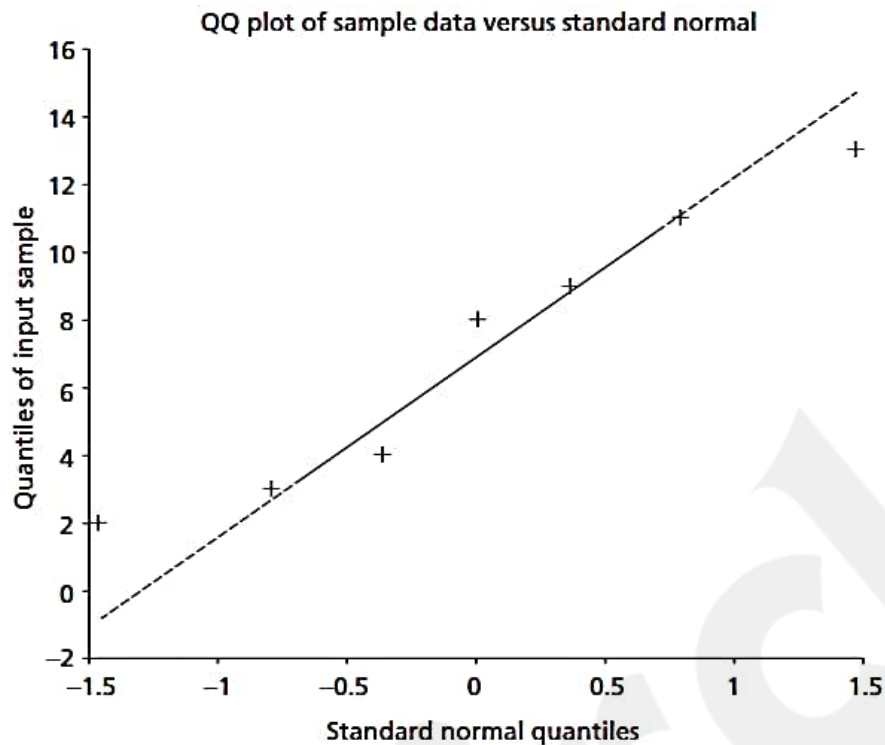


Figure 2.10: Normal Q-Q Plot

Ideally, the points fall along the reference line (45 Degree) if the data follows normal distribution. If the deviation is more, then there is greater evidence that the datasets follow some different distribution, that is, other than the normal distribution shape. In such a case, careful analysis of the statistical investigations should be carried out before interpretation.

This skewness, kurtosis, mean absolute deviation and coefficient of variation help in assessing the univariate data.